```
> restart:
> Odev:=proc(f,P,X,Y) local i,j,n, g,d1,XP, YP,TV, dXY,dYX,
  LieXY,LXf;
  n:= nops(X):

  print("Girilen Fonksiyon f" =f);
  print("Girilen nokta P"=P);
  print("Girilen vektor alani X" =X);
  print("Girilen vektor alani Y"=Y);


  d1:=seq(x[k]=P[k],k=1..n):
  XP:=[seq(subs(d1,X[i]),i=1..n)]:
  print("X(P)"=XP);

  YP:=[seq(subs(d1,Y[i]),i=1..n)]:
  print("Y(P)"=YP);

  TV:=[seq(50*XP[i]+51*YP[i],i=1..n)]:
  print("Tanjant Vektor: 50.X(P)+51.Y(P)"=TV);
  dXY:=[]:

  for j from 1 to n do
    for i from 1 to n do
       g[i]:=diff(Y[j],x[i]):
    od:
     dXY:=[op(dXY),simplify(sum(X[m]*g[m],m=1..n))]:
    od:
     print("D(X,Y) Kovaryant Turev"=dXY);
  dYX:=[]:


  for j from 1 to n do
     for i from 1 to n do
        g[i]:=diff(X[j],x[i]):
     od:
     dYX:=[op(dYX), simplify(sum(Y[m]*g[m], m=1..n) )]:
  od:

  print("D(Y,X) kovaryant turevi"=dYX);

  LieXY:=[seq(simplify(dXY[i]-dYX[i]), i=1..n)]:
  print("[X,Y] Lie Operatoru"= LieXY);

  for i from 1 to n do
  g[i]:= diff(f,x[i]):
```

```
od:
LXf:= simplify( sum(X[m]*g[m], m=1..n)):
print("LXf lie Turevi"= LXf);
print("LXY lie Turevi = [X,Y]"= LieXY);
end:
> #ORNEK1
> f:=x[1]*x[2]+3*x[2]^2+x[3]^2+5:
> P:=[-4,1,2]:
> X:=[x[3]^3+x[2]^4,3*x[1]^2*x[2],2*x[1]^3*x[2]]:
> Y:=[x[2]*x[3],2*x[1]^2,x[1]^3]:
> Odev(f,P,X,Y);
```

$$\text{"Girilen Fonksiyon f"} = x_1 x_2 + 3 x_2^2 + x_3^2 + 5$$

$$\text{"Girilen nokta P"} = [-4, 1, 2]$$

$$\text{"Girilen vektor alani X"} = [x_3^3 + x_2^4, 3 x_1^2 x_2, 2 x_1^3 x_2]$$

$$\text{"Girilen vektor alani Y"} = [x_2 x_3, 2 x_1^2, x_1^3]$$

$$\text{"X(P)"} = [9, 48, -128]$$

$$\text{"Y(P)"} = [2, 32, -64]$$

$$\text{"Tanjant Vektor: 50.X(P)+51.Y(P)"} = [552, 4032, -9664]$$

$$\text{"D(X,Y) Kovaryant Turev"} = [3 x_1^2 x_2 x_3 + 2 x_1^3 x_2^2, 4 (x_3^3 + x_2^4) x_1, 3 (x_3^3 + x_2^4) x_1^2]$$

$$\text{"D(Y,X) kovaryant turevi"} = [8 x_1^2 x_2^3 + 3 x_1^3 x_3^2, 6 x_2^2 x_3 x_1 + 6 x_1^4, 6 x_2^2 x_3 x_1^2 + 4 x_1^5]$$

$$\text{"[X,Y] Lie Operatoru"} = [3 x_1^2 x_2 x_3 + 2 x_1^3 x_2^2 - 8 x_1^2 x_2^3 - 3 x_1^3 x_3^2,$$
$$4 x_1 x_3^3 + 4 x_1 x_2^4 - 6 x_2^2 x_3 x_1 - 6 x_1^4, 3 x_1^2 x_3^3 + 3 x_1^2 x_2^4 - 6 x_2^2 x_3 x_1^2 - 4 x_1^5]$$

$$\text{"LXf lie Turevi"} = x_2 x_3^3 + x_2^5 + 3 x_1^3 x_2 + 18 x_1^2 x_2^2 + 4 x_1^3 x_2 x_3$$

$$\text{"LXY lie Turevi = [X,Y]"} = [3 x_1^2 x_2 x_3 + 2 x_1^3 x_2^2 - 8 x_1^2 x_2^3 - 3 x_1^3 x_3^2,$$
$$4 x_1 x_3^3 + 4 x_1 x_2^4 - 6 x_2^2 x_3 x_1 - 6 x_1^4, 3 x_1^2 x_3^3 + 3 x_1^2 x_2^4 - 6 x_2^2 x_3 x_1^2 - 4 x_1^5]$$

```
> #ORNEK2
> f:=2*x[1]*x[2]+x[3]^2:
> P:=[1,-1,2]:
> X:=[x[1]^2,x[2]^2,x[3]^2]:
> Y:=[x[1]*x[3],x[2]*x[3],x[1]*x[2]]:
> Odev(f,P,X,Y);
```

$$\text{"Girilen Fonksiyon f"} = 2 x_1 x_2 + x_3^2$$

$$\text{"Girilen nokta P"} = [1, -1, 2]$$

$$\text{"Girilen vektor alani X"} = [x_1^2, x_2^2, x_3^2]$$

$$\text{"Girilen vektor alani Y"} = [x_1 x_3, x_2 x_3, x_1 x_2]$$

$$\text{"X(P)"} = [1, 1, 4]$$

"Y(P)" = [2, -2, -1]

"Tanjant Vektor: 50.X(P)+51.Y(P)" = [152, -52, 149]

"D(X,Y) Kovaryant Turev" = $[x_1^2 x_3 + x_3^2 x_1, x_2^2 x_3 + x_3^2 x_2, x_1^2 x_2 + x_2^2 x_1]$

"D(Y,X) kovaryant turevi" = $[2 x_1^2 x_3, 2 x_2^2 x_3, 2 x_1 x_2 x_3]$

"[X,Y] Lie Operatoru" = $[-x_1^2 x_3 + x_3^2 x_1, -x_2^2 x_3 + x_3^2 x_2, x_1^2 x_2 + x_2^2 x_1 - 2 x_1 x_2 x_3]$

"LXf lie Turevi" = $2 x_1^2 x_2 + 2 x_2^2 x_1 + 2 x_3^3$

"LXY lie Turevi = [X,Y]" = $[-x_1^2 x_3 + x_3^2 x_1, -x_2^2 x_3 + x_3^2 x_2, x_1^2 x_2 + x_2^2 x_1 - 2 x_1 x_2 x_3]$

>